OCN 479-001

Dr. Phil Bresnahan



Today



Prep for tour of Ocean Instrumentation Lab



Lab: Learn git vs. GitHub and their combined capabilities



Homework: Prepare to give a short (3-5 mins) presentation next week on your data

Presentation guidelines

- Team up with one or two other people (last week's lab teammates or new) and pick your favorite dataset on https://cormp.org/. Give a short (3–5 minutes) presentation in class next week on the following:
  - What the parameter(s) is(are)
  - What they look like (e.g., time-series plot (screenshot))
  - How they were measured (look up sensors that measure your parameter)
  - What studies it would likely be used for
  - Important technical specifications (look up on website and/or product datasheets)
  - Why you picked it
- Your presentation should be no more than two or three slides which you will submit on Canvas (via the Assignment post)
- You will also deliver presentation in class next week or following week, time-dependent

What's the point?

- Familiarize yourself with cutting-edge coastal technologies and datasets
- Connect technology with applications
- Begin conversations about key metrics (e.g., accuracy, precision, stability, response time, endurance/lifetime, servicing intervals, technical needs)

Software and Services for Smart Coasts

- GitHub:
  - Online/website (with some desktop tools to be discussed later).
  - Create a free account if you don't already have one at github.com!
  - Copilot genAI coding tool is free for students, and we're a Microsoft campus w/ a license to Copilot
- VS Code:
  - A desktop app/software used for writing/reading code or any kind of text file.
  - Probably the most important software tool we'll use this semester!
- Particle:
  - Particle is the company that makes the hardware (primarily Argons) we're using
  - Workbench or Particle Workbench is a special set of software tools that run inside VS Code to allow us to compile (huh?) and flash (what?) code to Particle Argons
  - Create a Particle account on their website (particle.io) also



Blink_LED.cpp firmware explanation

- Particle devices run C++, a “low level” compiled application development language
- **You don’t need to learn much about writing new C++ code for this class, but finding and integrating useful code chunks will be essential**
- Particle devices have a script running on them that handles stuff “under the hood” and which calls your code as well
- It starts with setup(), which it executes once
- It then runs loop(), which it runs in a, well, loop!
- I recommend Copilot and w3schools to learn more ☺

```
/*
 * Project myProject
 * Author: Your Name
 * Date:
 * For comprehensive documentation and examples, please visit:
 * https://docs.particle.io/firmware/best-practices/firmware-template/
 */

// Include Particle Device OS APIs
#include "Particle.h"

// We define MY_LED to be the LED that we want to blink.
//
// In this tutorial, we're using the blue D7 LED (next to D7 on the Photon
// and Electron, and next to the USB connector on the Argon and Boron).
const pin_t MY_LED = D7;

// The following line is optional, but recommended in most firmware. It
// allows your code to run before the cloud is connected. In this case,
// it will begin blinking almost immediately instead of waiting until
// breathing cyan,
SYSTEM_THREAD(ENABLED);
SYSTEM_MODE(SEMI_AUTOMATIC);

// Random note to show git tools

// The setup() method is called once when the device boots.
void setup()
{
  // Particle.disconnect();
  // WiFi.off();
    // In order to set a pin, you must tell Device OS that the pin is
    // an OUTPUT pin. This is often done from setup() since you only need
    // to do it once.
    pinMode(MY_LED, OUTPUT);
}

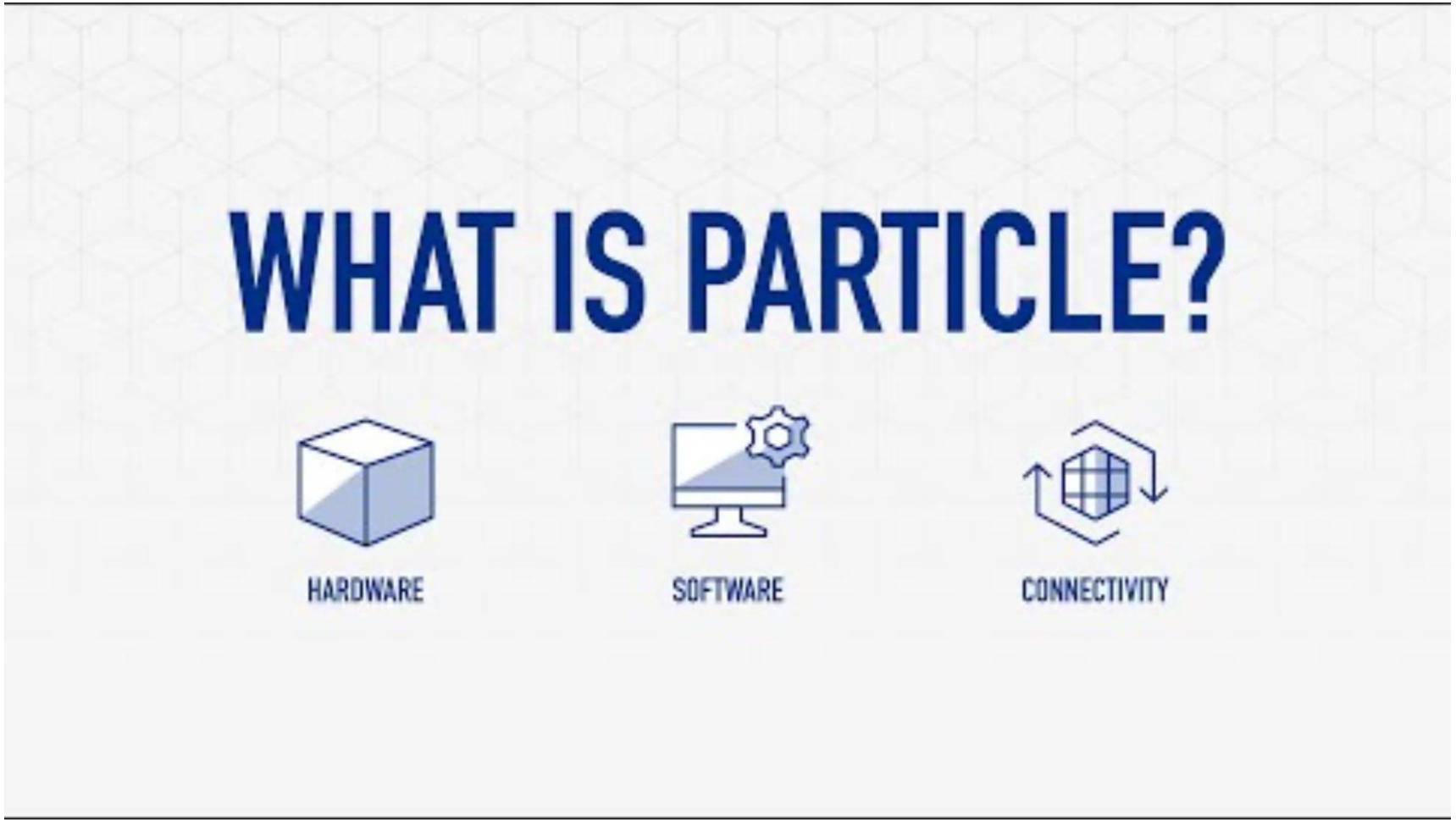
// The loop() method is called frequently.
void loop()
{
    // Turn on the LED
    digitalWrite(MY_LED, HIGH);

    // Leave it on for one second
    delay(3s);

    // Turn it off
    digitalWrite(MY_LED, LOW);

    // Wait one more second
    delay(1s);

    // And repeat!
}
```


WHAT IS PARTICLE?
HARDWARE
SOFTWARE
CONNECTIVITY

Learn about some of the other cool projects that use Particle on their website and YouTube channel…



Particle Stories: Shared Technologies launches smart scooters, concept to…

Particle • 3.6K views • 6 years ago

We spoke to the team at Shared Technologies to learn how they used Particle to improve their scooter sharing service.



Riding the IoT wave with Smartfin, an Electron connected Surfboard Fin

Particle • 9.2K views • 7 years ago

We sat down with Phil Bresnahan from the Smartfin project at Scripps Institute of Oceanography to find out how the Electron's cellular…

GitHub

Introduced briefly in OCN 350; to be used extensively in OCN 479

Reinforces “open science and engineering” concepts

Enables collaborative coding with a little more code stability than something like google docs. Collaborate with…

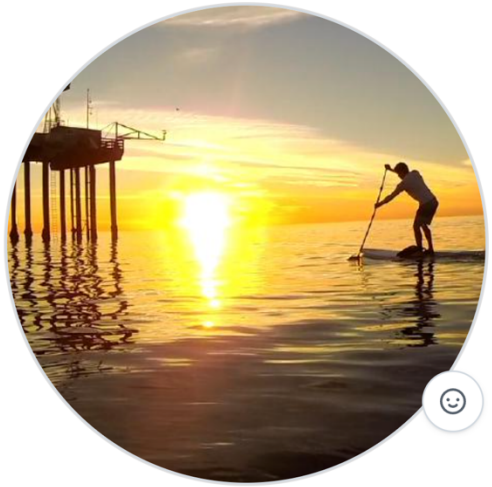
Teammates	Rest of this class	Future classes	People anywhere/anywhen in the world

Including links to GitHub repos on resumes/CVs will be very attractive to employers/graduate schools!

SUPScientist

Type / to search

Overview | Repositories 68 | Projects | Packages | Stars 51



Phil Bresnahan

SUPScientist · he/him

I'm a coastal oceanographer using data science and software/hardware engineering to learn more about the health of our natural resources.

Edit profile

17 followers · **8** following

Wilmington, North Carolina

SUPScientist / README.md

Hello! I'm Phil Bresnahan. Welcome to my GitHub profile

- 👷 Work: I'm currently working on the repos that you see lower on my profile here, the COAST-Lab org, and the TinyCamML org
- 📚 Background: I started as a chemical/environmental engineer, pivoted to chemical oceanography, and am now a coastal water resources specialist (the intersection of environmental engineering and oceanography)
- 🌊 Water always: When I'm not building tools to learn about the coast or teaching/learning more, I'm usually in or near the water! I love surfing, open water rowing, stand-up and prone paddling, and cycling/running near the water.

Where I Work 🏄

Here's a snippet from our lab's homepage, which I built using GitHub Pages (you can find the source code here).

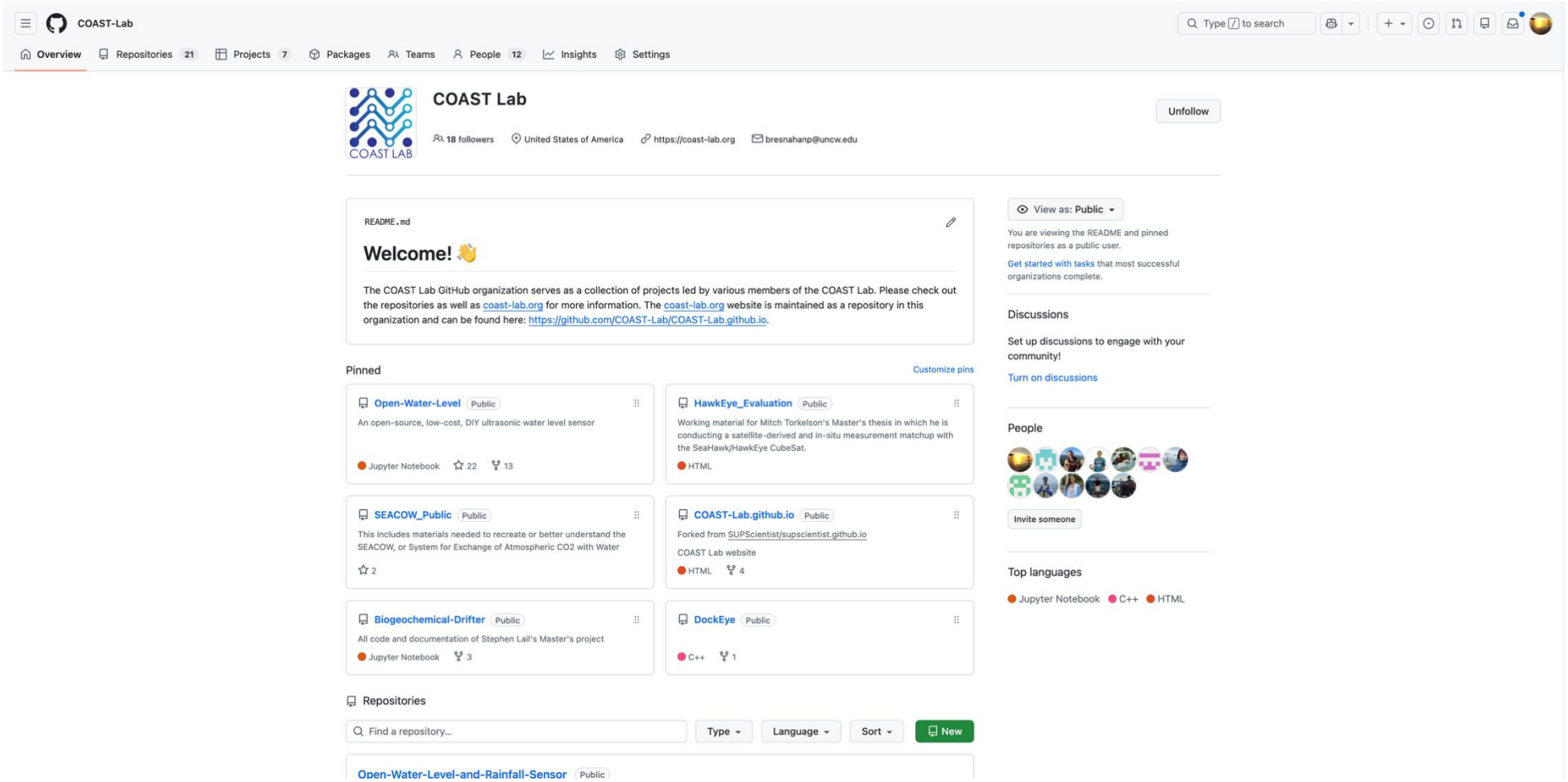
The Coastal Ocean Applied Science & Technology Lab seeks to find answers to and develop and communicate solutions for our most pressing environmental questions and problems. PI Phil Bresnahan and the COAST Lab are situated at the University of North Carolina Wilmington and affiliated with the Department of Earth and Ocean Sciences and Center for Marine Science. We focus on coastal water resources and resilience, and we employ a combination of ocean and environmental engineering and applied science approaches.

In short, we seek to contribute to Smart Coasts by:

- inventing and deploying cutting-edge connected sensors and platforms,
- engaging with coastal communities in the study and real-time dissemination of critical information, and
- working toward improved ocean, climate, and coastal literacy for students of all ages.

Our tools include (1) existing sensors and our own inventions paired with buoys, profilers, ships, and autonomous systems for *in-*

https://github.com/supscientist



https://github.com/COAST-Lab

That’s a lot of new terminology…

questions?

GitHub exercise

- Read this page: https://github.com/SUPScientist/Smart-Coasts/tree/main/Class-02-OILab-Tour
- After reading everything, complete at least the required and ideally the optional steps